

Theory:

- ① We know that convolution of a function and a filter or kernel is equivalent to multiplying the Fourier transforms of the function and filter.
- ② From Computer Vision, it is known that the Fourier transform of the 2D spatial image ~~represents~~ represents the image in the frequency domain. Also, the low frequencies of the Fourier-transformed image directly relate to the homogeneous regions (no or low intensity changes) of the spatial image. Likewise, the high frequencies directly relate to the edges or noise (high intensity changes).
- ③ Edge Detection:

The Laplacian (second-order derivative) of the spatial function can be used to find edges with "zero-crossings". The Laplacian can be Fourier-transformed with the following formulas:

$$\text{Laplacian: } \nabla^2 f(x, y) = f''(x) + f''(y)$$

Fourier Transform of Laplacian:

$$\mathcal{F}\{\nabla^2 f(x, y)\} = -4\pi(u^2 + v^2)F(u, v)$$

Note: $F(u,v) = \mathcal{F}\{f(x,y)\}$
 $H(u,v) = \mathcal{F}\{h(x,y)\}$

So, the edges of an image can be found by applying a high-pass filter to remove the low frequencies of the image. This can be done in the frequency domain by multiplying the Fourier transform of the image ^{uplication} with the Fourier transform of the high-pass filter.

$$\mathcal{F}\{\nabla^2 f(x,y) * h(x,y)\} =$$

$$\mathcal{F}\{f''(x) + f''(y)\} \cdot H(u,v)$$

~~Note that $\nabla^2 f(x,y) = f''(x) + f''(y)$~~

$$\mathcal{F}\{-4\pi^2(u^2 + v^2)F(u,v)\} \cdot H(u,v)$$

④ Region Segmentation:

Similar to edge detection, the regions of an image to be segmented can be found by applying a low-pass filter to remove the high frequencies of the image. This can be done in the frequency domain by multiplying the Fourier transform of the image with the Fourier transform of the low-pass filter.

$$\mathcal{F}\{f(x,y) * h(x,y)\} = F(u,v) \cdot H(u,v)$$